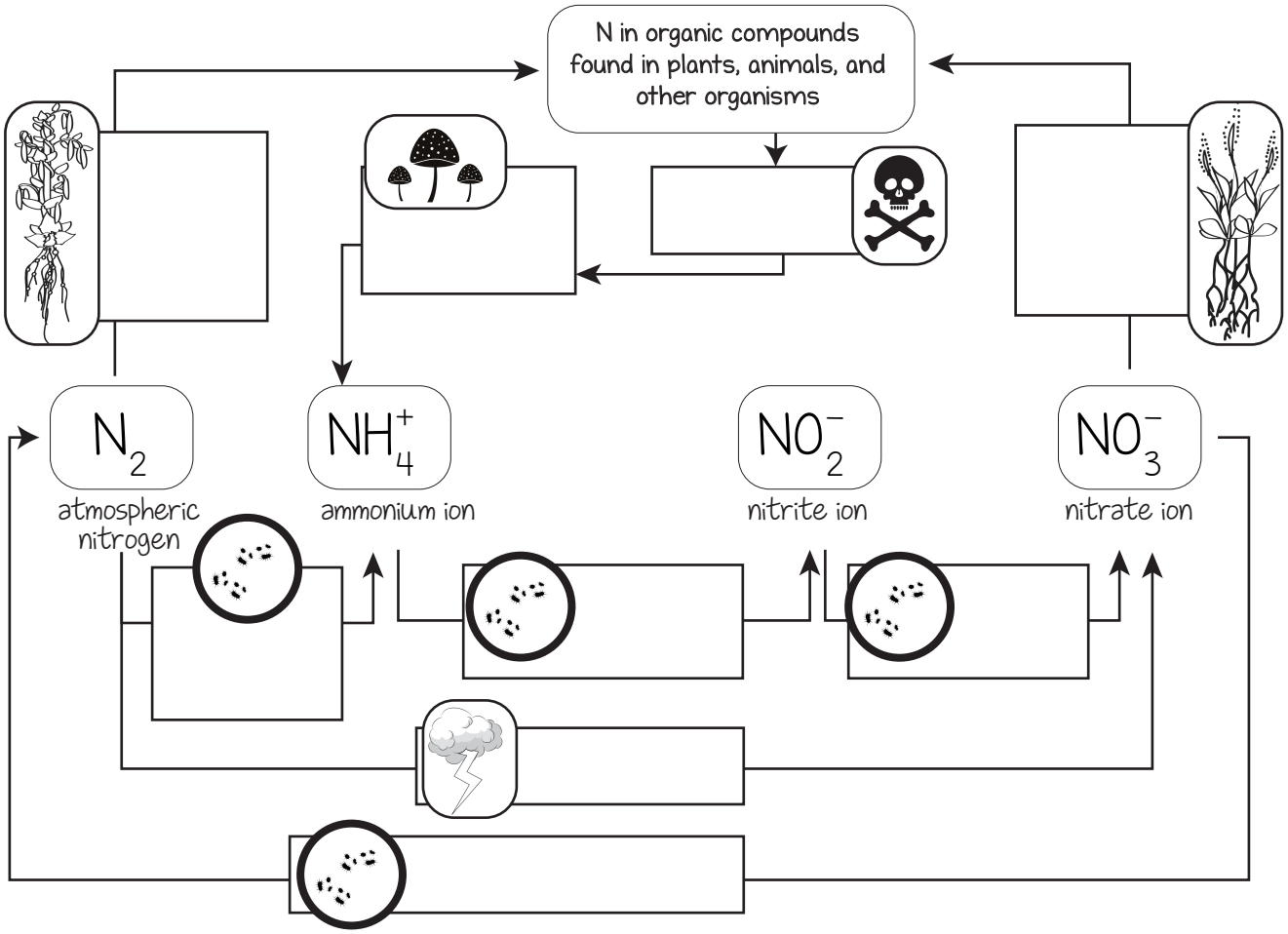
Name: Class: Date:

Nitrogen Cycle

Using the wordbank below, fill in the boxes to describe the players in the nitrogen cycle



1. decomposition by fungi and bacteria
2. lightning
3. most plant uptake
4. dead organisms and waste
5. nitrogen fixing bacteria in legume root nodules
6. nitrogen fixing bacteria in soil
7. denitrifying bacteria
8. ammonium oxidizing bacteria
9. nitrite oxidizing bacteria

Nitrogen is found in abundance in the atmosphere as N2, but this gas is not biologically available to organisms until it is "fixed" into ammonium ions. Nitrogen available to organisms to use is often the limiting ingredient, the limiting factor that determines plant productivity in an ecosystem.

10. When farmers add fertilizer to soil, what are they normally adding to the soil? What form(s) of nitrogen? (look it up if you don't know!)

Much of the nitrogen that humans add to the soil ends up in our streams, rivers, and the ocean. An increase of nitrogen in the waters can change the pH of the water, it can cause lower levels of oxygen in the water, and it can affect which organisms live there.

11. One consequence of increased aquatic nitrogen is toxic algal blooms. What is an algal bloom?

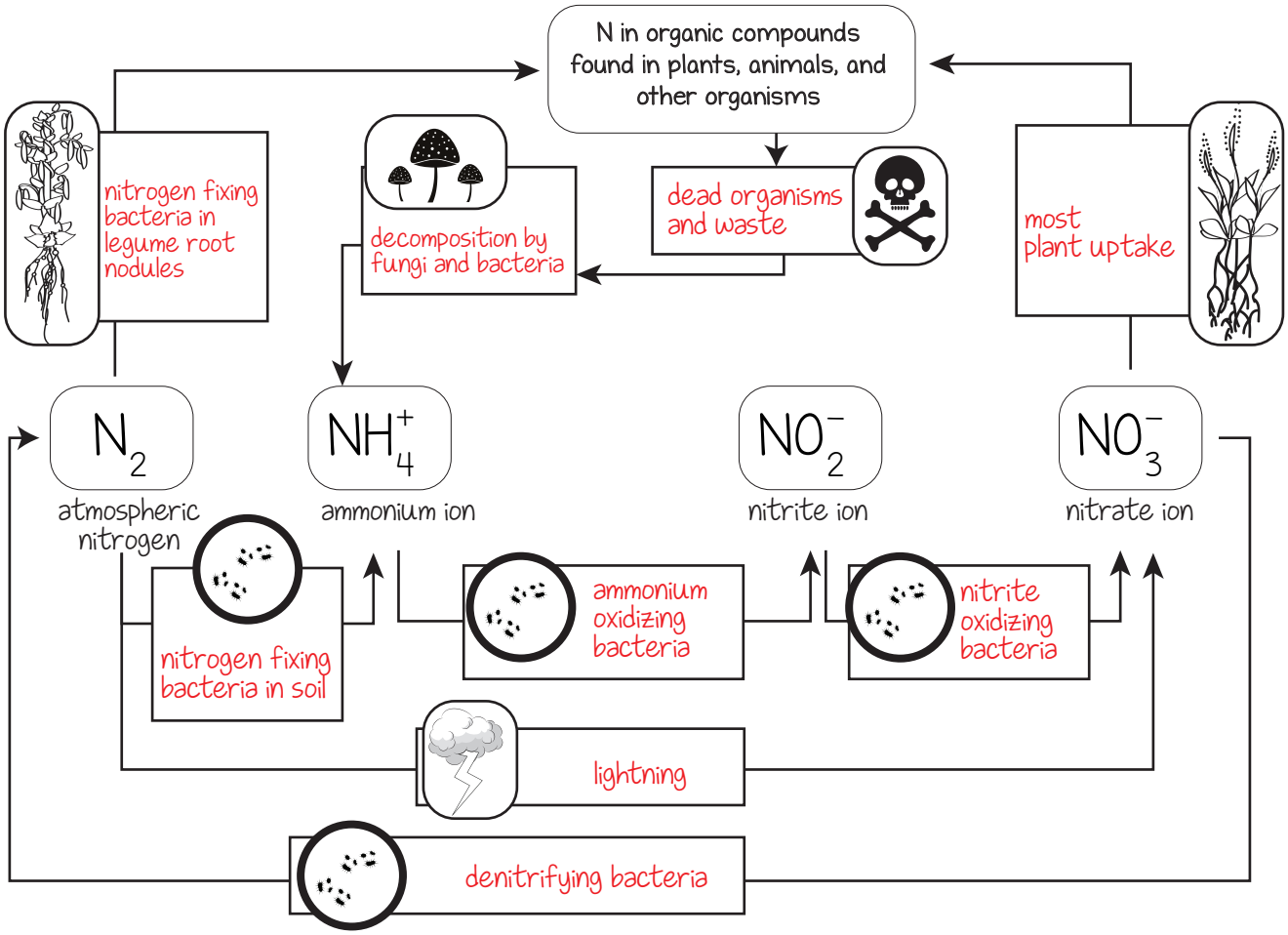
12. How do algal blooms affect other life in the surrounding area?

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Name: Answer Key Class: Date:

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10. When farmers add fertilizer to soil, what are they normally adding to the soil? What form(s) of nitrogen? (look it up if you don't know!)

Common fertilizer nitrogen components are ammonia, ammonium nitrate, and urea. (Urea is often preferable because its non-explosive!)

Much of the nitrogen that humans add to the soil ends up in our streams, rivers, and the ocean. An increase of nitrogen in the waters can change the pH of the water, it can cause lower levels of oxygen in the water, and it can affect which organisms live there.

11. One consequence of increased aquatic nitrogen is toxic algal blooms. What is an algal bloom?

Algal blooms are a large increase of algae in an area. Often the water changes color because of the dramatic increase of algae.

12. How do algal blooms affect other life in the surrounding area?

Algae can grow very quickly in high nutrient water (containing nitrogen-rich runoff) but they don't live long. What then happens is a high concentration of dead organic material fills the water. As the organic material from the algae decays, dissolved oxygen in the water is consumed. With less oxygen in the water, many animals and plants may die.

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